Quantum Imaginary-Time Evolution (QITE)

Study notes and benchmark plan against Quantum Krylov

Prajjwal

Started 2026-10-08

Last built: October 12, 2026

Contents

1 QITE from first principles **1**
- 1.1 Why imaginary time finds the ground state . . . 1
- 1.2 Why $e^{-\tau H}$ cannot simply be a circuit . . . 3
- 1.3 The QITE algorithm . . . 3
- 1.4 Variational QITE (McLachlan) and relations . . . 5
- 1.5 Cost, conditioning and pitfalls . . . 5
- 1.6 QITE versus Quantum Krylov . . . 5

2 Numerical check on H_2 **6**

3 Benchmark design on Quantinuum hardware **7**

4 Quick reference and references **8**

5 Correction: the origin of the QITE plateau on H_2 **9**

1 QITE from first principles

2026-10-08

Question 1. Prepare very good notes on Quantum Imaginary-Time Evolution (QITE): why imaginary time finds the ground state, why $e^{-\tau H}$ cannot be run directly as a circuit, how QITE replaces it by a unitary $e^{-i\Delta\tau A}$ with A determined from measured expectation values, how it compares with Quantum Krylov, and how to benchmark both on Quantinuum hardware for the molecular ground-state energy (e.g. H_2).

Reading guide. Sections 1.1 and 1.2 are the physics. Section 1.3 is the algorithm and its derivation, the part to master. Section 1.5 is cost and pitfalls, Section 1.6 the comparison with Krylov. The numerical check on H_2 is in Section 2 and the experimental design in Section 3. Throughout, $H = \sum_m h_m P_m$ is a qubit Hamiltonian with Pauli strings P_m, $\hbar = 1$, and energies in Hartree.

1.1 Why imaginary time finds the ground state

Let $H|E_n\rangle = E_n|E_n\rangle$ with $E_0 < E_1 \le E_2 \le \cdots$ and write the initial state as $|\psi_0\rangle = \sum_n c_n|E_n\rangle$. Replace real time e^{-iHt} by imaginary time $t = -i\tau$:

$$|\psi(\tau)\rangle = \frac{e^{-\tau H}|\psi_0\rangle}{\|e^{-\tau H}|\psi_0\rangle\|}, \qquad e^{-\tau H}|\psi_0\rangle = e^{-\tau E_0}\Big[c_0|E_0\rangle + \sum_{n>0} c_n e^{-\tau(E_n-E_0)}|E_n\rangle\Big]. \tag{1}$$

Proposition 1.1 (Convergence). *If $c_0 \neq 0$ and $\Delta = E_1 - E_0 > 0$ is the gap, then $|\psi(\tau)\rangle \to |E_0\rangle$ and*

$$E(\tau) - E_0 \;\le\; \frac{\sum_{n>0} |c_n|^2 (E_n - E_0)\, e^{-2\tau(E_n - E_0)}}{|c_0|^2} \;=\; O(e^{-2\tau\Delta}), \qquad E(\tau) = \frac{\langle\psi(\tau)|H|\psi(\tau)\rangle}{\langle\psi(\tau)|\psi(\tau)\rangle}. \tag{2}$$

Proof. $E(\tau) - E_0 = \sum_n |c_n|^2 e^{-2\tau(E_n - E_0)}(E_n - E_0) / \sum_n |c_n|^2 e^{-2\tau(E_n - E_0)}$. The denominator is at least $|c_0|^2$, giving the bound. □

Remark. Two things control the rate: the gap Δ (a small gap needs a long τ) and the overlap $|c_0|^2$ (the prefactor). Hartree–Fock typically has $|c_0|^2 \approx 0.9+$, so the prefactor is benign and the gap is the real cost driver. Only states in the same symmetry sector as $|\psi_0\rangle$ matter: for H_2 in Section 2 the relevant gap is that of the 2×2 sector containing HF (1.632 Ha), not the global gap (0.613 Ha).

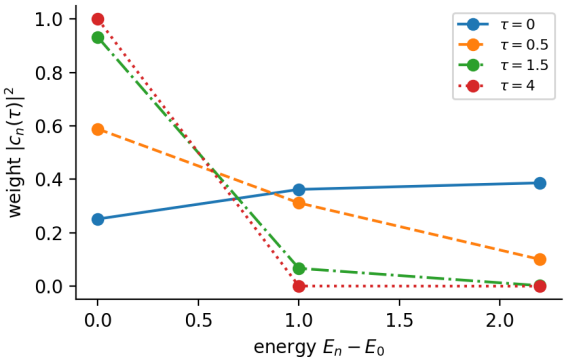


Figure 1: Imaginary-time evolution as a filter: weights $|c_n(\tau)|^2$ on a toy three-level spectrum. Excited weight decays as $e^{-2\tau(E_n - E_0)}$.

The normalised flow and the energy-descent law

The normalised state obeys a nonlinear equation. Differentiating the quotient above,

$$\frac{\mathrm{d}}{\mathrm{d}\tau}|\psi\rangle = -(H - E(\tau))|\psi\rangle. \tag{3}$$

The $E(\tau)$ term is exactly what keeps the norm fixed; the physical content is $-H|\psi\rangle$ projected orthogonal to $|\psi\rangle$.

Proposition 1.2 (Energy descent). *Along Equation (3),* $\dfrac{\mathrm{d}E}{\mathrm{d}\tau} = -2\,\mathrm{Var}_\psi(H) = -2(\langle H^2\rangle - \langle H\rangle^2) \le 0$, *with equality iff $|\psi\rangle$ is an eigenstate.*

Proof. $\frac{\mathrm{d}E}{\mathrm{d}\tau} = 2\,\mathrm{Re}\langle\psi|H|\dot\psi\rangle = -2\,\mathrm{Re}\langle\psi|H(H-E)|\psi\rangle = -2(\langle H^2\rangle - E\langle H\rangle) = -2\,\mathrm{Var}_\psi(H)$. □

Remark (Diagnostic). Proposition 1.2 is exact for ideal ITE. For QITE the measured E should be (nearly) monotone and $\mathrm{Var}_\psi(H) \to 0$. An increasing energy signals approximation error (domain too small, $\Delta\tau$ too large, ill-conditioned S) or shot noise/hardware noise. The variance $\mathrm{Var}_\psi(H)$ is also a rigorous eigenstate certificate: some eigenvalue lies within $\sqrt{\mathrm{Var}_\psi(H)}$ of $E(\tau)$.

Remark (Power iteration). $e^{-\tau H}$ is the continuous analogue of power iteration with the filter $f(H) = e^{-\tau H}$ (equivalently $(\lambda_{\max} - H)^k$). This is why QITE and Krylov methods are cousins: both are polynomial/filter approximations to a spectral projector.

1.2 Why $e^{-\tau H}$ cannot simply be a circuit

A circuit implements a unitary, $U^\dagger U = I$. But $e^{-\tau H}$ is Hermitian positive definite with $(e^{-\tau H})^\dagger e^{-\tau H} = e^{-2\tau H} \neq I$ for any $\tau \neq 0$ unless $H \propto I$. It rescales each eigencomponent by $e^{-\tau E_n}$, so it changes both norm and relative weights. (The statement "$|e^{-\tau E_n}| < 1$" only holds if $E_n > 0$; the point is that the factors are *not of modulus one*.) The state after normalisation is physical, but the map is not unitary.

There are three families of workarounds:

1. *Unitary surrogate (QITE, Motta et al.)*: find a Hermitian A with $e^{-i\Delta\tau A}|\psi\rangle \approx e^{-\Delta\tau H}|\psi\rangle/\sqrt{c}$ *on the current state.*
2. *Variational (VarQITE, McLachlan)*: parametrise $|\psi(\vec{\theta})\rangle$ and propagate $\vec{\theta}$ by an ODE (Section 1.4).
3. *Probabilistic/ancilla (block-encoding, LCU)*: embed $e^{-\tau H}$ in a larger unitary and post-select; succeeds with probability $\sim \|e^{-\tau H}|\psi\rangle\|^2$, which decays exponentially in τ. Not NISQ-friendly.

1.3 The QITE algorithm

Step 1: Trotterise the Hamiltonian

Write $H = \sum_{m=1}^{M} h_m$ with each h_m a *local* Pauli term (coefficient times Pauli string, possibly grouped). To first order in $\Delta\tau$,

$$e^{-\Delta\tau H} = \prod_{m=1}^{M} e^{-\Delta\tau h_m} + O(\Delta\tau^2). \tag{4}$$

Each factor $e^{-\Delta\tau h_m}$ acts on few qubits, which is what makes a local unitary surrogate plausible. For $h_m = \alpha_m P_m$ with $P_m^2 = I$, $e^{-\Delta\tau h_m} = \cosh(\Delta\tau\alpha_m)I - \sinh(\Delta\tau\alpha_m)P_m$.

Step 2: One local step

Given the current normalised state $|\psi\rangle$, define the target

$$|\psi'\rangle = \frac{e^{-\Delta\tau h_m}|\psi\rangle}{\sqrt{c_m}}, \qquad c_m = \langle\psi|e^{-2\Delta\tau h_m}|\psi\rangle = 1 - 2\Delta\tau\langle h_m\rangle + O(\Delta\tau^2). \tag{5}$$

We want a Hermitian $A_m = \sum_{\mu\in\mathcal{D}_m} a_\mu\sigma_\mu$, with σ_μ Pauli strings in a domain $\mathcal{D}_m$ (typically all strings on the qubits within distance D of the support of h_m), such that $e^{-i\Delta\tau A_m}|\psi\rangle \approx |\psi'\rangle$. Expanding to first order, $e^{-i\Delta\tau A_m}|\psi\rangle \approx |\psi\rangle - i\Delta\tau A_m|\psi\rangle$, so we need

$$-iA_m|\psi\rangle \approx |d\rangle \coloneqq \frac{|\psi'\rangle - |\psi\rangle}{\Delta\tau}. \tag{6}$$

Proposition 1.3 (Linear system for $\vec{a}$)**.** *Minimising $\|-iA_m|\psi\rangle - |d\rangle\|^2$ over real $\vec{a}$ gives*

$$\boxed{S\vec{a} = \vec{b}, \qquad S_{\mu\nu} = \mathrm{Re}\langle\psi|\sigma_\mu\sigma_\nu|\psi\rangle, \qquad b_\mu = -\,\mathrm{Im}\langle\psi|\sigma_\mu|d\rangle} \tag{7}$$

Proof. The cost is $f(\vec{a}) = \|\sum_\nu a_\nu(-i\sigma_\nu|\psi\rangle) - |d\rangle\|^2$. Setting $\frac{\partial f}{\partial a_\mu} = 0$ gives $\sum_\nu a_\nu\,\mathrm{Re}\langle -i\sigma_\mu\psi| - i\sigma_\nu\psi\rangle = \mathrm{Re}\langle -i\sigma_\mu\psi|d\rangle$. Now $\langle\sigma_\mu\psi|\sigma_\nu\psi\rangle = \langle\psi|\sigma_\mu\sigma_\nu|\psi\rangle$ and $\mathrm{Re}(i\langle\psi|\sigma_\mu|d\rangle) = -\,\mathrm{Im}\langle\psi|\sigma_\mu|d\rangle$. □

Everything is a measurement. $S_{\mu\nu}$ is the expectation of the Pauli string $\sigma_\mu\sigma_\nu$ (a Pauli up to a phase), estimated by measuring on the current circuit $U_k\cdots U_1|\psi_0\rangle$. For $\vec{b}$, substitute $|d\rangle$:

$$b_\mu = -\frac{1}{\Delta\tau}\,\mathrm{Im}\left[\tfrac{1}{\sqrt{c_m}}\langle\psi|\sigma_\mu e^{-\Delta\tau h_m}|\psi\rangle - \langle\psi|\sigma_\mu|\psi\rangle\right], \qquad e^{-\Delta\tau h_m} = \cosh(\Delta\tau\alpha_m) - \sinh(\Delta\tau\alpha_m)P_m, \tag{8}$$

and $\langle\psi|\sigma_\mu|\psi\rangle$ is real (Hermitian Pauli), so only the $\sigma_\mu P_m$ term survives:

$$b_\mu = \frac{\sinh(\Delta\tau\alpha_m)}{\Delta\tau\sqrt{c_m}}\,\mathrm{Im}\langle\psi|\sigma_\mu P_m|\psi\rangle. \tag{9}$$

Since $\sigma_\mu P_m$ is $\pm i$ times a Hermitian Pauli whenever σ_μ, P_m anticommute (and Hermitian when they commute), b_μ is nonzero only for *anticommuting* pairs: each is a single Pauli expectation. No ancillas, no Hadamard tests, only basis rotations and computational-basis measurement. This is the key practical feature.

Real structure. For a real Hamiltonian and a real initial state (H_2, HF) the state stays real, $\langle\psi|\sigma_\mu|\psi\rangle = 0$ when σ_μ has an even number of Y, and S restricted to strings with an *odd* number of Y is the relevant block. This halves the domain.

Step 3: Implement the unitary

The step operator is $e^{-i\Delta\tau A_m}$ with $A_m = \sum_\mu a_\mu\sigma_\mu$; the strings in A_m do not commute, so one Trotterises again, $e^{-i\Delta\tau A_m} \approx \prod_\mu e^{-i\Delta\tau a_\mu\sigma_\mu}$, each factor a Pauli rotation (CNOT ladder around one R_z, or native ZZ-type phase gates on trapped-ion hardware). Then set $|\psi\rangle \leftarrow e^{-i\Delta\tau A_m}|\psi\rangle$ and move to $m+1$.

Algorithm (one pass over M terms = one imaginary-time step $\Delta\tau$).

1. Prepare $|\psi\rangle = U_k\cdots U_1|\psi_0\rangle$ (all previous step unitaries).
2. For $m = 1,\ldots,M$:
   (a) Choose the domain $\mathcal{D}_m$ (size D, $d_m = |\mathcal{D}_m|$ Pauli strings).
   (b) Measure $\langle h_m\rangle$ (for c_m), all $S_{\mu\nu}$, all b_μ.
   (c) Solve $S\vec{a} = \vec{b}$ (regularised, least squares).
   (d) Append $U_{k+1} = e^{-i\Delta\tau\sum_\mu a_\mu\sigma_\mu}$ to the circuit.
3. Measure $E = \langle H\rangle$ (and $\mathrm{Var}\,H$); stop when converged, else repeat.

Remark (Correction to a common simplification). It is tempting to describe QITE as "repeatedly executing relatively shallow circuits $U_1,\ldots,U_K$". Each U_j is shallow, but the state at step k is $U_k\cdots U_1|\psi_0\rangle$: the circuit depth grows linearly with the number of steps, and one cannot reuse the previous state without re-preparing it. Depth per step is small; total depth is not. This is the main difference from a fixed-depth VQE ansatz and is what the benchmark must measure honestly (Section 1.5).

Locality and the size of the domain

Why should a local A_m exist at all? If the state has finite correlation length ξ, then $e^{-\Delta\tau h_m}|\psi\rangle$ differs from $|\psi\rangle$ only by a quasi-local perturbation, and Motta et al. show the approximation error decays with the domain size roughly as $e^{-D/\xi}$, whereas the number of strings grows as 4^D (per dimension-dependent volume). For a general molecule, where interactions are not geometrically local after a fermion-to-qubit map, the useful domain can be all 4^{N_q} strings, which is exponential. For H_2 (2 qubits) the full domain is just 16 (or 6 odd-Y) strings, which is why it is an easy test case and also why it is not representative of scaling.

Imaginary time ↔ unitary: what is exact

With $\mathcal{D}$ the full Pauli group the equation $-iA|\psi\rangle = |d\rangle$ has an exact solution (for any two non-orthogonal states there is a Hermitian generator mapping one to the other), so the only remaining errors are the *first-order expansion* in $\Delta\tau$, the Trotter splitting Equation (4) and the *conditioning* of S. Truncating $\mathcal{D}$ adds a systematic error on top.

1.4 Variational QITE (McLachlan) and relations

Take an ansatz $|\psi(\vec{\theta})\rangle$ and impose the McLachlan principle $\delta\|(\partial_\tau + H - E)|\psi\rangle\| = 0$, giving

$$\sum_j M_{ij}\dot{\theta}_j = -C_i, \qquad M_{ij} = \mathrm{Re}\langle\partial_i\psi|\partial_j\psi\rangle, \quad C_i = \tfrac{1}{2}\partial_i E. \tag{10}$$

M is the quantum geometric tensor (Fubini–Study metric up to normalisation and the $\langle\partial_i\psi|\psi\rangle$ terms) so VarQITE is *natural gradient descent* on $E(\vec{\theta})$. Differences from QITE: fixed circuit depth (set by the ansatz) but an *ansatz-expressibility* error and possible local minima; QITE has no ansatz, but its depth grows. If one writes the QITE generator A as a parameter of an ever-growing ansatz, QITE is the same linear system with a growing parameter set.

1.5 Cost, conditioning and pitfalls

Per step. With M terms and domain size d per term the number of distinct Pauli expectation values is $O(Md^2)$ for S and $O(Md)$ for $\vec{b}$ (many repeated). Each must be estimated to precision ϵ, costing $O(1/\epsilon^2)$ shots, grouped by qubit-wise commuting measurement bases.

Circuit growth. After N steps there are $\sim NM$ unitaries each with up to d Pauli rotations:

$$\text{depth} \;\sim\; N \cdot M \cdot d \cdot (\text{depth of one Pauli rotation}), \qquad N_{2q} \;\sim\; NMd\,\bar{w}, \tag{11}$$

$\bar{w}$ being the mean weight of the strings. The number of circuits executed to reach step N is $\sim \sum_{k\le N}(\text{measurement settings} \cdot \text{shots})$, i.e. quadratic in N if circuit lengths grow linearly and one pays for the whole prefix at every step.

Ill-conditioning. Near convergence, $\sigma_\mu|\psi\rangle$ become nearly linearly dependent (indeed $|\psi\rangle \to$ eigenstate, so $A|\psi\rangle \to$ small, and many strings give the same vector): S becomes singular. With shot noise, $\vec{a} = S^{-1}\vec{b}$ amplifies the noise by $\kappa(S)$. Standard remedies: Tikhonov $S+\epsilon I$, truncated SVD / pseudo-inverse, or a larger $\Delta\tau$ so $\vec{b}$ is not tiny relative to the noise. This is the QITE analogue of the overlap-matrix thresholding in Krylov.

Summary table of pitfalls.

Source	Effect	Mitigation
$\Delta\tau$ too large	Trotter and first-order error, energy plateau above E_0	Halve $\Delta\tau$; second-order splitting
$\Delta\tau$ too small	Many steps, deep circuit, noise accumulates	Choose $\Delta\tau \sim 0.1$–$0.5/\|H\|$ scale; adaptive
Small domain	Systematic bias, may stall	Grow D; check against exact for small N_q
Ill-conditioned S	Noise amplification, energy drifts up	Regularise, SVD cut-off, more shots
Hardware noise	Mixed state, E above the ideal curve	Error mitigation; compare on identical noise model
Symmetry sector	Wrong state if $\|\psi_0\rangle$ has no c_0	Initialise in the right (N, S_z) sector

1.6 QITE versus Quantum Krylov

Quantum Krylov (QKSD / filter diagonalisation) prepares states $|\phi_j\rangle = e^{-iHt_j}|\psi_0\rangle$ ($j = 0, \ldots, K-1$; commonly $t_j = j\,\delta t$), measures

$$H_{jk} = \langle\phi_j|H|\phi_k\rangle, \qquad S_{jk} = \langle\phi_j|\phi_k\rangle, \tag{12}$$

by Hadamard-type or time-shifted-overlap circuits, and solves the generalised eigenproblem $HC = ESC$. For real-time evolution with equally spaced times both matrices are Toeplitz-like: $S_{jk} = s(t_k - t_j)$, so only $\sim 2K$ distinct overlaps are required.

	Quantum Krylov	QITE
Quantum part	Overlaps $\langle\psi_0\|e^{-iHt}\|\psi_0\rangle$ and $\langle\psi_0\|He^{-iHt}\|\psi_0\rangle$ (Hadamard test or equivalents)	Pauli expectation values $S_{\mu\nu}, b_\mu$ at every step
Classical part	One small generalised eigenproblem	A least-squares solve per term per step
Circuit per run	Time evolution e^{-iHt} (Trotter), depth $\propto t$, *fixed per K*	Prefix grows with step number
Error source	Trotter error in e^{-iHt}; S ill-conditioning when δt small or K large	Domain truncation, first-order $\Delta\tau$, S ill-conditioning
Convergence	Subspace error $\sim$ Chebyshev-like in K; independent of a step schedule	$e^{-2\tau\Delta}$ at best
Noise robustness	Overlap noise enters the generalised eigenproblem; thresholding required	Noise enters each step and is *accumulated* in the state
Qubits	N_q (+1 ancilla for Hadamard test)	N_q

Remark (Why they are cousins). Real-time Krylov with $t_j = j\,\delta t$ spans $\{p(e^{-iH\delta t})|\psi_0\rangle\}$ over polynomials of degree $< K$; QITE applies one specific filter $e^{-\tau H}$. Krylov *optimises* the filter (Rayleigh–Ritz picks the best combination in the span), so for a given number of applications of the propagator it can converge faster than a fixed filter, but needs the classical post-processing and robust S. QITE has no such optimisation, but needs no overlap circuits.

2 Numerical check on H_2

2026-10-08

Question 2. Implement QITE for the 2-qubit H_2 Hamiltonian, compare with exact ITE and a small Quantum Krylov baseline, and quantify the effect of shot noise.

Setup. H_2, STO-3G, $R = 0.735$ Å, parity mapping with two-qubit reduction (the standard 2-qubit Hamiltonian; nuclear repulsion 0.7200 Ha excluded):

$$H = -1.0524\,II + 0.3979\,IZ - 0.3979\,ZI - 0.0113\,ZZ + 0.1809\,XX. \tag{13}$$

Exact diagonalisation: $E_{\mathrm{FCI}} = -1.857275$ Ha; Hartree–Fock (the basis state $|01\rangle$) -1.836968 Ha, i.e. a correlation energy of 20.3 mHa; $|c_0|^2 = 0.9877$. Code: `code/qite_h2.py`, run in `aq-env`. QITE uses the full odd-Y domain $\{IY, XY, YI, YX, YZ, ZY\}$, $\Delta\tau = 0.1$, 60 steps, one first-order Trotter pass per step, regularisation 10^{-6}. Krylov uses real time $\delta t = 0.5$ with an exact propagator (this isolates the subspace error; Trotter error is not included). Shot noise is emulated by adding Gaussian noise of size $1/\sqrt{N_{\mathrm{shots}}}$ to every entry of S and $\vec{b}$ (a crude model, independent per entry and ignoring the shot cost of grouping).

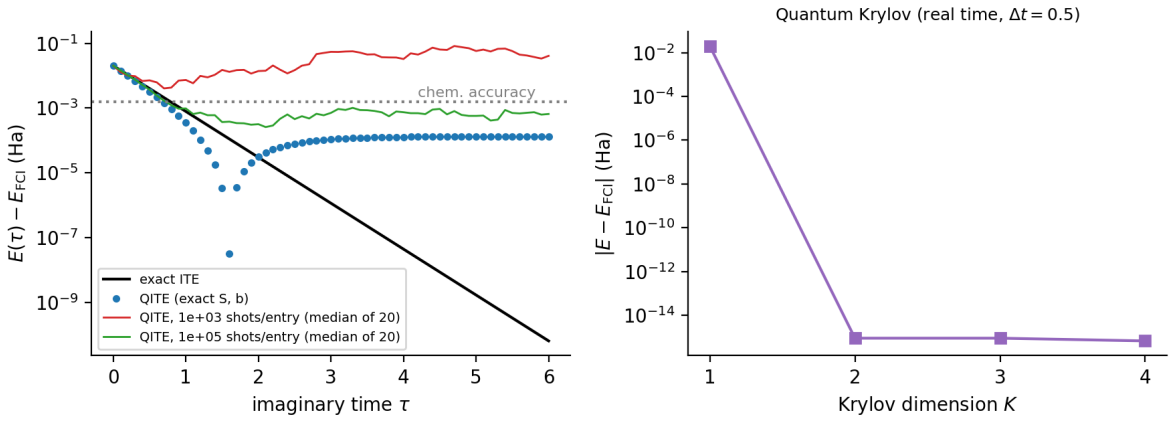


Figure 2: Left: energy error versus imaginary time for exact ITE, QITE with exactly evaluated $S, \vec{b}$, and QITE with shot-noisy $S, \vec{b}$ (median of 20 seeds). Right: Quantum Krylov versus subspace dimension.

Quantity	Value
Steps to chemical accuracy (1.6 mHa), exact ITE / QITE	8 / 7
Final error, exact ITE ($\tau = 6$)	$\sim 10^{-10}$ Ha
Best / final error, QITE (exact $S, \vec{b}$)	3×10^{-8} ($\tau = 1.6$) / 1.3×10^{-4} Ha
Plateau error, QITE, 10^3 shots/entry (mean over 20 seeds)	0.15 Ha
Plateau error, QITE, 10^5 shots/entry	2.0×10^{-3} Ha
Quantum Krylov error, $K = 1, 2, 3, 4$	2.0×10^{-2}, $< 10^{-14}$, $< 10^{-14}$, $< 10^{-14}$ Ha

Conclusions.

1. With exact $S, \vec{b}$ QITE reproduces exact ITE down to chemical accuracy in a few steps, confirming the derivation in Proposition 1.3.
2. The QITE energy is not monotone: it reaches an error of 3×10^{-8} Ha at $\tau = 1.6$ and then *drifts back up* to a plateau at $\approx 1.3 \times 10^{-4}$ Ha, whereas exact ITE keeps decaying. This is the singularity of S near the ground state discussed in Section 1.5, interacting with the regularisation and the first-order step; it violates Proposition 1.2 and is exactly the symptom the variance diagnostic should flag. Energies stayed above E_{FCI} (variational), as they must.
3. Shot noise is far more damaging than the algorithm error: at 10^3 shots per matrix entry the result is worse than the starting HF point (error 0.15 Ha > HF error 0.020 Ha) because noisy S is inverted. At 10^5 shots the plateau is roughly chemical accuracy. This is the quantitative form of "QITE needs many measurements".
4. Quantum Krylov is exact at $K = 2$ here only because the HF state lives in a 2×2 symmetry sector, so any 2-dimensional Krylov space spans it. H_2 cannot discriminate the two methods: it is a test of correctness, not of scaling. A discriminating benchmark needs ≥ 4 qubits (e.g. LiH active space, H_4) where the sector is large.

3 Benchmark design on Quantinuum hardware

2026-10-08

Question 3. Under identical Quantinuum resource constraints, how do QITE and Quantum Krylov trade accuracy, circuit depth, number of measurements and robustness?

Metrics (record all, per method and per budget point).

Metric	Definition
Accuracy	$\lvert E - E_{\mathrm{FCI}}\rvert$ in Ha and mHa; also $\mathrm{Var}_\psi(H)$ where available
Shots	total N_{shots} summed over *all* circuits, including those for S, H-matrix, $\vec{b}$
Circuits	number of distinct circuit executions N_{circ} (after grouping)
Depth	maximum circuit depth D over the run (QITE: last step; Krylov: deepest e^{-iHt})
2-qubit gates	N_{2q} per circuit and in total
Qubits	N_q (+ancilla)
Wall time	T; for hardware also queue time and compile time reported separately

Experiments.
A. Ideal simulation Noiseless statevector for both methods vs. E_{FCI}; checks the code and Trotter/domain errors alone.
B. Noisy simulation One *shared* noise model (depolarising $2q$ rate, readout error, finite shots) for both; separates algorithmic from hardware error.
C. Hardware Both on the same Quantinuum device/emulator, same date, same compiler settings.
D. Budget-normalised Fix N_{shots} and/or N_{2q}; compare accuracy at equal cost.
E. Scaling curve Vary K (Krylov) and number of QITE steps N; plot $\lvert E - E_{\mathrm{FCI}}\rvert$ against hardware cost (shots, or N_{2q} total). This is the central figure.

Fair-comparison rules.
1. Same initial state (HF), same qubit mapping/tapering, same Hamiltonian grouping.
2. Count *every* circuit run, including repeated prefixes in QITE and Hadamard-test ancillas in Krylov.
3. Tune hyperparameters ($\Delta\tau$, domain, regularisation; δt, K, threshold) on the noisy *simulator*, then freeze them before hardware.
4. Report seeds / repeated runs: with shot noise a single curve is not evidence (cf. the 20-seed median above).
5. Always include the plain baselines: HF energy and VQE (or fixed-ansatz) at the same budget.

Hypotheses to test (not results).
- QITE has shallow per-step circuits but total depth growing with step count; it may win at small budgets and lose when the prefix grows.
- Krylov needs deeper e^{-iHt} with fewer total repetitions but is sensitive to overlap noise; the generalised eigenproblem may fail when S is noisy.
- Which wins on Quantinuum (high-fidelity, all-to-all two-qubit gates, mid-circuit measurement) is empirical. Trapped-ion connectivity favours QITE's non-local Pauli rotations, since no SWAP overhead is incurred, which removes one common penalty for large domains.

Open items. [TODO: Confirm with the colleague which Krylov variant (real-time vs. filtered, Hadamard test vs. overlap-only) and which time grid t_j.] [TODO: Get the actual molecule, basis, active space, qubit mapping and Quantinuum target (device vs. emulator) before fixing the noise model.] [TODO: Extend `qite_h2.py` to $N_q \geq 4$ (e.g. LiH active space) with a locality-truncated domain, plus a Trotterised Krylov baseline.]

4 Quick reference and references

2026-10-08

Formula sheet.

$$\text{ITE:}\quad |\psi(\tau)\rangle \propto e^{-\tau H}|\psi_0\rangle,\quad |\dot{\psi}\rangle = -(H-E)|\psi\rangle,\quad \dot{E} = -2\,\mathrm{Var}_\psi\, H$$
$$\text{Trotter:}\quad e^{-\Delta\tau H} \approx \textstyle\prod_m e^{-\Delta\tau h_m}$$

$$\begin{aligned}
\text{Surrogate:} &\quad e^{-\Delta\tau h_m}|\psi\rangle/\sqrt{c_m} \approx e^{-i\Delta\tau A_m}|\psi\rangle,\ A_m = \textstyle\sum_\mu a_\mu \sigma_\mu \\
\text{Linear system:} &\quad S_{\mu\nu} = \mathrm{Re}\langle\sigma_\mu\sigma_\nu\rangle,\ b_\mu = \tfrac{\sinh(\Delta\tau\alpha_m)}{\Delta\tau\sqrt{c_m}}\,\mathrm{Im}\langle\sigma_\mu P_m\rangle \\
\text{Krylov:} &\quad HC = ESC,\ S_{jk} = \langle\phi_j|\phi_k\rangle,\ |\phi_j\rangle = e^{-iHt_j}|\psi_0\rangle
\end{aligned}$$

Self-test questions.
1. Why is $b_\mu = 0$ when σ_μ commutes with P_m?
2. Why does S become singular as $|\psi\rangle \to |E_0\rangle$, and what does regularisation trade away?
3. Show that the McLachlan equation with a rotation ansatz reproduces Equation (7).
4. Which gap controls convergence for a state in a given symmetry sector?

References (quoted from memory; check details before citing).

- M. Motta et al., "Determining eigenstates and thermal states on a quantum computer using quantum imaginary time evolution", *Nature Physics* **16**, 205 (2020).
- S. McArdle et al., "Variational ansatz-based quantum simulation of imaginary time evolution", *npj Quantum Information* **5**, 75 (2019).
- R. M. Parrish and P. L. McMahon, "Quantum filter diagonalization", arXiv:1909.08925 (2019).
- N. H. Stair, R. Huang, F. A. Evangelista, "A multireference quantum Krylov algorithm for strongly correlated electrons", *J. Chem. Theory Comput.* **16**, 2236 (2020).
- E. N. Epperly, L. Lin, Y. Nakatsukasa, "A theory of quantum subspace diagonalization", *SIAM J. Matrix Anal. Appl.* **43**, 1263 (2022).

5 Correction: the origin of the QITE plateau on H_2 2026-10-12

Question 4. Is the $\approx 10^{-4}$ Ha plateau in the two-qubit H_2 QITE run really caused by the singularity of S, as concluded in the numerical-check entry above?

No. That explanation was asserted, not tested. Re-running `code/qite_h2.py` with exact $S, \vec{b}$:

$\Delta\tau$	QITE final error (Ha)	Product of exact term filters, no QITE fit (Ha)
0.1	1.33×10^{-4}	1.33×10^{-4}
0.05	3.24×10^{-5}	3.24×10^{-5}
0.025	7.99×10^{-6}	7.99×10^{-6}

The plateau scales like $\Delta\tau^2$ and is identical with and without the QITE fit, and changing the regularisation from 10^{-6} to 10^{-12} leaves it unchanged. It is therefore the Trotter-splitting error (the terms XX and IZ, ZI do not commute, so the fixed point of $\prod_m e^{-\Delta\tau h_m}$ is not exactly the ground state), not ill-conditioning of S. The *shot-noise* conclusion stands, and the fix there is regularisation larger than the noise level: with 10^5 shots the mean error falls from 2.9×10^{-2} (ridge 10^{-6}) to 1.5×10^{-4} Ha (ridge 10^{-2}). The corrected discussion is in `qite-complete-guide.tex`.